Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12	(a)			(enthalpy change) when 1 mol (of substance) is burned (1) in excess oxygen (1)	2			2		
	(b)	(i)	I	n(propanol) = $\frac{0.14}{60}$ = 0.00233		1		1	1	1
			II	Δ*T* = 12.5 (1) energy = *mc*Δ*T* = 100 × 4.18 × 12.5 = 5225 J (1) Δ*H* = $-\frac{5.225}{0.00233}$ = – 2242 kJ mol^{-1} (1)		3		3	3	2
		(ii)		credit possible for agree / disagree award (1) for any of following agree – because the temperature rise will be greater (making the percentage error lower) agree – because greater mass of fuel burned (making the percentage error lower) disagree – because more heat will be lost to the surroundings (making Δ*T* value less accurate)			1	1		1